

Aniket Mishra

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EXPERIENCE

Software Engineer

May 2025 – Present

Total Life

Vancouver, BC (Remote)

- Developed a concurrent job queue in **Go** using **goroutines**, **buffered channels**, and a **configurable worker pool** to process patient health data **asynchronously**, **improving throughput by 25%** compared to the previous sequential processing system.
- Automated the **end-to-end billing and CRM lifecycle** by integrating **HubSpot** and **Stripe** through custom middleware that synchronized patient records and processed payment webhooks, **reducing manual data entry and invoicing errors by 6.33%**.
- Architected a **custom Pub/Sub messaging framework** to route patient communications across SMS and email (via **Twilio**), **Slack APIs**, and in-app notifications, centralizing delivery workflows and **reducing manual handling by 31%**.

Software Engineer

September 2024 – April 2025

vrCAVE

Edmonton, Alberta

- Optimized IMMDevice state management in **C++** with multithreading, eliminating a 5-second frame stall and **reducing average frame processing time by 7.25%**.
- Refactored legacy code using enums, structs, and delegates, resulting in **improved code maintainability** and a robust hint system for multiplayer VR environments.
- Designed a custom latent function for state syncing across Meta Quest headsets, **reducing multiplayer latency spikes by 20%** verified via Unreal Engine profiling.

Software Developer Intern

May 2023 – December 2023

G2V Optics

Edmonton, Alberta

- Developed a **solar tracking** feature for Small Area LED Solar Simulator for Aerospace ground testing that calculates **solar irradiance** based on a user's specific longitude, latitude, elevation, and time.
- Created a **Bash script** to automate Raspberry Pi setup and **Apache environment configuration**, reducing a 30 minute manual process to under 10 minutes.

Software Developer Intern

January 2022 – August 2022

AI Shading

Calgary, Alberta

- Replaced REST polling with **WebSockets** for real-time sensor updates from Tuya SDK and Open Weather APIs, **reducing dashboard latency from 250ms to 150ms**.
- Used **Datadog** for logging and monitoring to track down error patterns, helping to identify and **fix recurring bugs** and **reducing runtime issues by 16%**.

PROJECTS

NotExo.earth | *NASA Kepler Dataset, Scikit-learn, React, Bun, Python, GCP, Docker*

(Website)

- Founded a volunteer-led **research group** applying machine learning to **open datasets from major space agencies** to advance accessible research in **exoplanet detection**, **planetary science**, and **climate analytics**.
- Used **Stacking Ensemble** and **Random Forest** models on the **NASA Kepler dataset** to classify exoplanets with high precision, and built an **interactive canvas-based star map** for real-time visualization of exoplanet coordinates.

Business Boardroom | *Gemini, LangGraph, GCP, Docker, Next.js, FastAPI, Python*

(Demo)

- Architected a **multi-agent** business simulation engine for the **Kaggle x Google GenAI Contest**, utilizing **LangGraph** and **Gemini** to orchestrate stateful, role-based dialogues (CEO, Strategist, Intern) for complex strategic decision-making.
- Deployed a **scalable GCP infrastructure via Cloud Run**, integrating a **FastAPI** backend with a **Next.js** frontend and a **Slack extension** to enable cross-platform agent interactions.

FissionMunk | *Python, NumPy, pytest, PyPI, Poetry, GitHub Actions*

(PyPI)

- Developed a **lightweight 2D physics open source Python library** designed to simulate nuclear fission reactor mechanics, enabling researchers and enthusiasts to make interactive simulations of nuclear reactors.
- Published on PyPI (**pip install fissionmunk**), ensuring easy installation. Implemented **CI/CD pipelines with 100% automated test coverage using pytest and GitHub Actions**, reducing integration bugs.
- Integrated WebSocket support to output each simulation step as JSON, enabling **real-time data streaming** and visualization in external applications.

EDUCATION

University of Alberta

Edmonton, Alberta

Bachelor of Science with Specialization in Computer Science — GPA 3.7

September 2019 – June 2024

- Dean's Honor Roll (2021-2024), International Student Scholarship

SKILLS

Languages/Frameworks: Go, Python, C, C++, Java, Next.js, React, Django, Flask, Svelte, JavaScript, HTML/CSS, FastAPI

Technologies: PostgreSQL, Git, Numpy, DataDog, Google Cloud Platform, Cloud Run, BigQuery, Firebase, Docker